Deetya Kamat

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Education

University of Washington – Paul G. Allen School of Computer Science & Engineering Seattle, WA
B.S. Computer Science: Data Science Option
Minors: Applied Mathematics, Business Administration *Expected June 2027*
Dean's List Recipient, Microsoft Endowed Scholarship (2024)
Relevant Coursework: Distributed Systems, Algorithms, Machine Learning, Systems Programming, Natural Language Processing, Data Structures

Experience

SingleStore - Software Engineering Intern Seattle, WA
July 2026 - September 2026 *Helios Dev: Analyst (AI & Compute Platform)*

- Owned end-to-end implementation of context version management for SingleStore's Analyst tool, enabling creators to create drafts, publish immutable releases, view version history, and roll back production context used by AI chat.
- Designed release/snapshot data models and production context resolution, implementing backend APIs across Aura Context Service and Nova Gateway in Go with database migration and backward-compatibility support.
- Integrated versioned context across backend and Creator Mode UI, implementing feature-flagged release history, publish/restore flows, feedback attribution, and migration plumbing for canary rollout.

Paul G. Allen School of Computer Science and Engineering - Research Assistant Seattle, WA
December 2025 - Present

- Engineered Python analysis pipelines to evaluate AlphaGenome sequence-to-function predictions across multimodal RNA-seq and DNase-seq genomic datasets.
- Benchmarked model prediction reliability across 500 PrediXcan genes, identifying variant- and tissue-level performance variances against individual-level ROSMAP brain RNA-seq data.

Projects

Google Drive Metadata Sync Service - *Python, SQLite, Google Drive API* January 2026 - February 2026

- Built a resumable, idempotent Google Drive metadata synchronization service with persistent SQLite state for file metadata, sync checkpoints, and job state.
- Implemented a fault-tolerant job runner with retry limits, checkpoint recovery, and automatic restart of interrupted jobs; added unit tests for synchronization correctness.

Multi-threaded Search Engine – Systems Programming Project - *C, C++, POSIX* July 2025 - August 2025

- Designed and implemented a scalable, multi-threaded query-based search engine with efficient file indexing and memory management.
- Optimized data structures and synchronization for concurrent search and dynamic memory management.

Spotify ML Recommender - *Python, Spotify API, ML* July 2025 - February 2026

- Built a personalized recommendation pipeline extracting listening history and metadata across playlists using Spotipy and pandas.
- Ranked candidate tracks via cosine similarity against the target user vector, caching artist metadata to optimize API request overhead.

Leadership & Activities

GEN1 @ UW – First-Generation Computing Organization Seattle, WA
Chair July 2026 - Present
Vice Chair September 2025 - June 2026
Student Relations Officer September 2024 - June 2025
Historian October 2023 - June 2024

- Supported organization-wide operations and programming as Vice Chair, leading major events including First-Generation Week and First-Gen Graduation and managing GEN1nterns.
- Created biweekly student newsletters achieving a 70% open rate and grew GEN1's Instagram following from 300 to 400 followers through student-focused outreach.

Technical Skills

Languages: Go, Python, Java, C++, C, JavaScript/TypeScript
Databases & Tools: MongoDB, SQLite, SQL, Linux/UNIX, Git, REST APIs, Jupyter
Frameworks & Libraries: React, Express.js, Pandas, NumPy, Scikit-learn, PyTorch, Vega-Lite